

Project: Smart Parking Platform

Version: 2.1

Date: 09.17.2026

Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

This document defines the initial requirements for the Smart Parking Platform developed by WE ARE \$oftware Corp. The system will help drivers locate, reserve, and pay for parking spaces in real time while providing parking operators with tools to manage their facilities and monitor occupancy. The platform will be available through web and mobile applications and will provide parking availability information, navigation assistance, and management tools for parking operators. The goal is to develop a project management plan for the creation of this system, including requirements gathering, stakeholder management, scheduling, budgeting, risk management, and team coordination.

1.2 Scope

The project will provide web and mobile applications that support parking searches, real-time availability, reservations, digital payments, navigation, notifications, and parking management. The project will also provide an administrative dashboard for parking operators to manage facilities and view occupancy information and reports. Physical parking garage construction, manufacturing parking hardware, and direct control of city traffic systems are out of scope.

2. Overall Description

- Users:

- Vehicle Owners
- Parking Facility Owners/Employees
- Finance and Billing Employees
- Third-Party Payment Providers
- City Transportation Employees
- Mobile Phone Users

- System Environment:

The Smart Parking Platform will consist of web-based and mobile applications connected to backend/API services and a database. The system will also integrate with external mapping/navigation and payment services.

- Constraints:

- The project has a fixed semester timeline.
- The project has an unlimited theoretical development budget.
- The system depends on third-party mapping, navigation, payment, and other APIs.
- The system must follow applicable security and privacy requirements.
- The system must consider applicable city ordinances and laws.
- Users must authenticate to access protected account and management functions.

- Assumptions:

- Users have internet access.
- Users have a compatible web browser or mobile device.
- Parking facilities provide accurate availability and occupancy information.
- Third-party mapping and payment services are available when required.
- Parking operators will provide and maintain accurate facility information.

3. Functional Requirements

FR1: The system shall allow users to register and log in securely.

FR2: The system shall allow users to search for available parking locations.

FR3: The system shall display parking locations and availability on an interactive map.

FR4: The system shall allow users to reserve available parking spaces.

FR5: The system shall process digital parking payments through a supported payment provider.

FR6: The system shall provide users with reservation history and receipts.

FR7: The system shall send users reservation-related notifications and alerts.

FR8: The system shall provide navigation assistance to selected parking facilities through an external mapping service.

FR9: The system shall allow parking operators to add and manage parking facilities.

FR10: The system shall allow parking operators to monitor parking occupancy.

FR11: The system shall provide occupancy and parking usage reports.

FR12: The system shall allow authorized users to manage relevant account and parking information.

Additional functional requirements will be identified through stakeholder analysis and requirements elicitation throughout the semester.

4. Non-Functional Requirements

NFR1: The system shall support at least 1,000 concurrent users.

NFR2: The system shall display parking availability updates within 1 second when current data is available.

NFR3: The system shall target 99.99% uptime for critical services.

NFR4: The system shall protect user account, vehicle, reservation, and payment-related information.

NFR5: The system shall support modern web browsers and supported mobile devices.

NFR6: The system shall be scalable to support additional users and parking facilities.

NFR7: The system shall maintain consistent reservation and payment information when transactions are completed successfully.

5. Use Cases

Use Case 1

Title: Reserve a Parking Space

Actor: Driver

Precondition: User is logged in and a parking facility has an available space.

Steps:

- User searches for a parking location.
- User selects an available parking facility.

- User selects the reservation date and time.
- System verifies parking availability.
- User confirms the reservation and completes payment.
- System displays the reservation confirmation.

Postcondition: The parking reservation is saved to the user's account.

Work Breakdown Structure (WBS)

1. Smart Parking Platform

1.1 User & Account Management

1.1.1 User Registration & Login

Account creation

Secure login

1.1.2 User Profile

Account information

Vehicle information

1.1.3 Authentication & Security

Authentication controls

Protected user data

1.2 Parking Search & Availability

1.2.1 Parking Location Search

Search available facilities

Filter parking locations

1.2.2 Real-Time Availability

Display available spaces

Update occupancy information

1.2.3 Interactive Parking Map

Display parking locations

Select parking facility

1.3 Reservations

1.3.1 Space Selection

Select parking facility

Select available space

1.3.2 Reservation Management

Select date and time

Confirm reservation

1.3.3 Reservation History

View past reservations

View reservation details

1.4 Payments

- 1.4.1 Payment Processing
 - Connect payment provider
 - Process parking payment
 - 1.4.2 Payment Confirmation
 - Confirm successful payment
 - Store transaction information
 - 1.4.3 Receipts
 - Generate digital receipt
 - View receipt history
- 1.5 Navigation & Notifications
 - 1.5.1 Navigation Integration
 - Connect mapping service
 - Provide directions to facility
 - 1.5.2 User Notifications
 - Reservation confirmation
 - Reservation alerts
- 1.6 Parking Operator Management
 - 1.6.1 Facility Management
 - Add parking facility
 - Update facility information
 - Remove facility
 - 1.6.2 Occupancy Management
 - Monitor occupancy
 - Update availability
 - 1.6.3 Operator Dashboard
 - View facility information
 - View parking activity
- 1.7 Reporting & Analytics
 - 1.7.1 Occupancy Reports
 - Generate occupancy reports
 - View occupancy trends
 - 1.7.2 Parking Usage Reports
 - Analyze parking usage
 - Review reservation activity
- 1.8 Testing & Deployment
 - 1.8.1 System Testing
 - Functional testing
 - Integration testing
 - 1.8.2 Security & Performance Testing
 - Security testing
 - Performance testing

1.8.3 Deployment

Deploy web application

Deploy mobile application

Deploy backend services

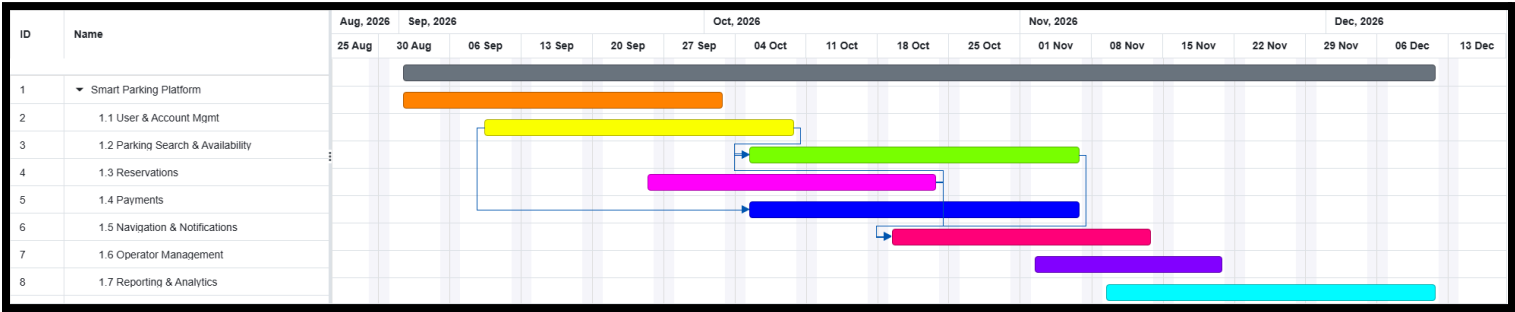
Project Timeline

WBS Component	Story Points	Risk
1.1 User & Account Mgmt	8	Moderate complexity; relies on secure auth standards
1.2 Parking Search & Availability	5	Depends on third-party map API reliability
1.3 Reservations	13	High complexity; requires state management & concurrency handling
1.4 Payments	8	Standard gateway integration but needs error/retry logic
1.5 Navigation & Notifications	3	Lightweight API calls; mostly configuration-driven
1.6 Operator Management	5	CRUD-heavy but straightforward UI/backend mapping
1.7 Reporting & Analytics	8	Requires data aggregation & charting libraries
1.8 Testing & Deployment	5	Cross-team coordination; environment setup overhead

WBS Component	Start	End	Duration	Dependencies
1.1 User & Account Mgmt	Sept Week 1	Oct Week 1	~5 wks	–
1.2 Parking Search & Availability	Sept Week 2	Oct Week 2	~5 wks	–
1.3 Reservations	Oct Week 1	Nov Week 1	~5 wks	1.2 (Finished)
1.4 Payments	Sept Week 4	Oct Week 3	~5 wks	1.4 (Started)
1.5 Navigation & Notifications	Oct Week 2	Nov Week 1	~5 wks	1.3 (Started)
1.6 Operator Management	Oct Week 3	Nov Week 2	~4 wks	1.3 (Finished~), 1.4 (Started)
1.7 Reporting &	Nov Week 1	Nov Week 3	~3 wks	Backend API Calling

Analytics				
1.8 Testing & Deployment	Nov Week 2	Dec Week 2	~5 wks	First working Dev component required to start testing

Gantt Chart



Milestones

Milestone	Target Date	Success Criteria
Requirements & WBS Baseline	Sept 7 (Week 1)	SRS v1.0 approved, WBS signed off by team leads
Core Search & Auth Complete	Oct 9 (Week 6)	Users can log in, search lots, view live availability
Reservation & Payment Flow Live	Nov 6 (Week 9)	End-to-end booking + payment confirmation working
Operator Dashboard & Reports Ready	Nov 27 (Week 10)	Facilities manageable, occupancy/reports viewable